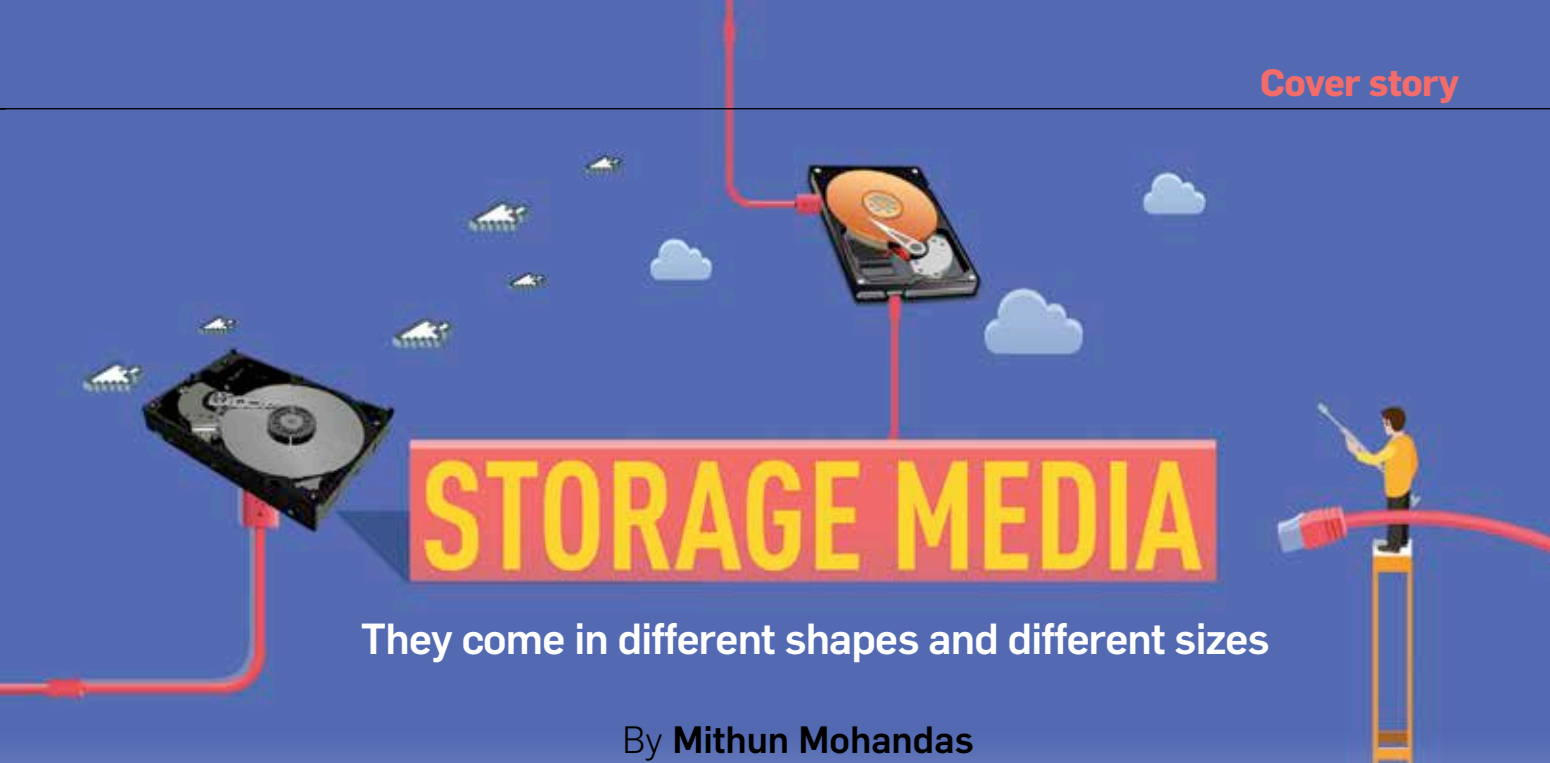




STORAGE MEDIA

They come in different shapes and different sizes

By **Mithun Mohandas**

We can never have enough of storage and our storage woes worsen with each passing year. There are vast improvements on the horizon in this segment with more manufacturers releasing PCIe based SSDs for the new M.2 standard. Moreover, NVMe has just been released and consumer models have started arriving. In light of these new technologies coming into the storage sector, the days of storage being the bottleneck in computers are soon to end. This clutter of new technologies also means that the user has more things to choose from while building a new computer, so here's how you'd go about testing these new storage devices.

Types

There are quite a few verticals by which you'd classify different storage products. There are SD cards, thumb drives, hard drives, SSDs, PCIe drives, M.2 drives and then there are devices which use some of the above drives to create storage devices with better data security/capacity/features. A NAS (Network Attached Storage) is one such device. Then you have to look at the interfaces through which these drives are connected to the computer. There is a plethora of options on this end as well. But we aren't going to worry about all of that, instead we're going to narrow down all of the above into categories that the

average consumer would be most interested in buying – thumb drives, SD cards, hard drives, SSDs and NAS devices.

Thumb Drives

Call 'em what you will, thumb drives, flash drives, pen drives or those little thingys that you stick into your computer and take data. Thumb drives are damn cheap and the number of SKUs available in the market outnumber the different types of sarees you'd find in a wedding store. Which means you do have the choice between going in for a lot of storage+poor transfer speed or less storage+great transfer speed or you can pay a lot and get the best of both worlds.

SD Cards

DSLR camera owners will be most interested in knowing if that Class 10 card is

really worth the money. After all, the class 10 certification only means that the minimum transfer speed will be 10MB/s but the upper ceiling can be anything. So shooting images in burst mode or shooting high quality video with less compression will require a good transfer speed.

Hard Drives

These are the traditional rotary hard drives which are still in high demand due to the low cost/GB ratio. This makes hard drives the staple primary storage in almost everything computer that is released. There are hybrid SSHDs which offer better performance but the cost/GB ratio is a teeny bit hit for mass adoption. Laptops with a really slim profile have made the switch from mechanical hard drives to SSDs.

SSDs

SSDs are way better than hard drives in terms of durability and speed but the current generation isn't long lasting if you are a heavy user. NAND chips only have a limited number of read/write cycles, a drawback that mechanical hard drives are free from.

NAS Devices

The benefits of a NAS are many, you get a lot of storage space, data protection, transfer speeds and reduced power consumption amongst others. You



An SSD is a must-have nowadays